ARCHITECTURAL PORTFOLIO

# Hanibal Ravandeh

## Senior Architectural Visualizer & BIM Specialist

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ABOUT

I work between ideas and evidence.

I am a Senior Architectural Visualizer and BIM Specialist with more than 13 years of experience working across architectural visualization, BIM, and design development.

My approach begins before the model.

I study the site, its constraints, its surroundings, and the conditions that can shape a project. Then I test ideas against those conditions rather than assuming that the first answer is the right one.

## Idea ↔ Site

I don't see an architectural idea and its site as separate steps.

That tension is where the work begins.

I develop alternatives seriously enough that each one can be judged on its own merits. Iteration is not about producing more options for the sake of it. It is about making the alternatives concrete enough to compare, test, and eventually defend a decision with evidence.

The principle behind this process is simple:

> I trust the first idea enough to fight for it — but not enough to stop testing it.

## How I Work

I use visualization and BIM not simply as production tools, but as ways of making architectural decisions visible.

RESIDENTIAL — VILLA RENOVATION (MERGING TWO EXISTING BUILDINGS)

# Villa Red Sun

Villa Red Sun merges two separate existing buildings into one coherent home, resolving a real conflict between cost-driven and quality-driven client priorities through seven tested design proposals.

Architectural Design, Visualization & BIM Documentation

3ds Max / V-Ray / Corona Renderer / AutoCAD / Revit

PROCESS

## 01 — CLIENT CHALLENGE

*What did the client actually need, and why was it hard?*

The client owned two separate residential buildings on the same property and wanted a single, functional home. Two family members held opposing priorities: the mother wanted to minimize demolition, construction work, and overall cost; the father wanted the best possible architecture, without financial limitations.

Site analysis — two buildings, one plot.

## 02 — SITE

*What was the physical starting condition?*

Two independent structures shared a single plot, each with its own orientation, daylight condition, and relationship to the site. Any merging strategy first had to be tested against how each building already related to light, privacy, and access.

## 03 — CONSTRAINTS

*What limits shaped every decision from here on?*

Budget and ambition pulled in opposite directions from within the same household. Connecting two buildings into one circulation system, without knowing in advance which financial scenario the client would ultimately accept, meant the design had to remain legible across multiple cost tiers rather than committing early to one.

## 04 — DESIGN THINKING

*How was the problem approached?*

Rather than proposing a single solution, seven design proposals were developed across three cost/quality tiers — A (low cost, minimum demolition), B (medium cost), and C (high quality, higher intervention) — before a final, complete architectural revision (D) was developed as the highest-quality synthesis.

## 05 — FINAL DECISION

*Why did this proposal win?*

Proposal D was selected. It fully integrated both buildings into a single, cohesive residence. It required the highest investment, but delivered the greatest improvement in functionality, spatial quality, and architectural identity — the criteria that mattered most once both family members saw the alternatives side by side.

## B-2

Medium cost

Master Plan

Floor Plan

## C-1

High quality, higher intervention

Master Plan

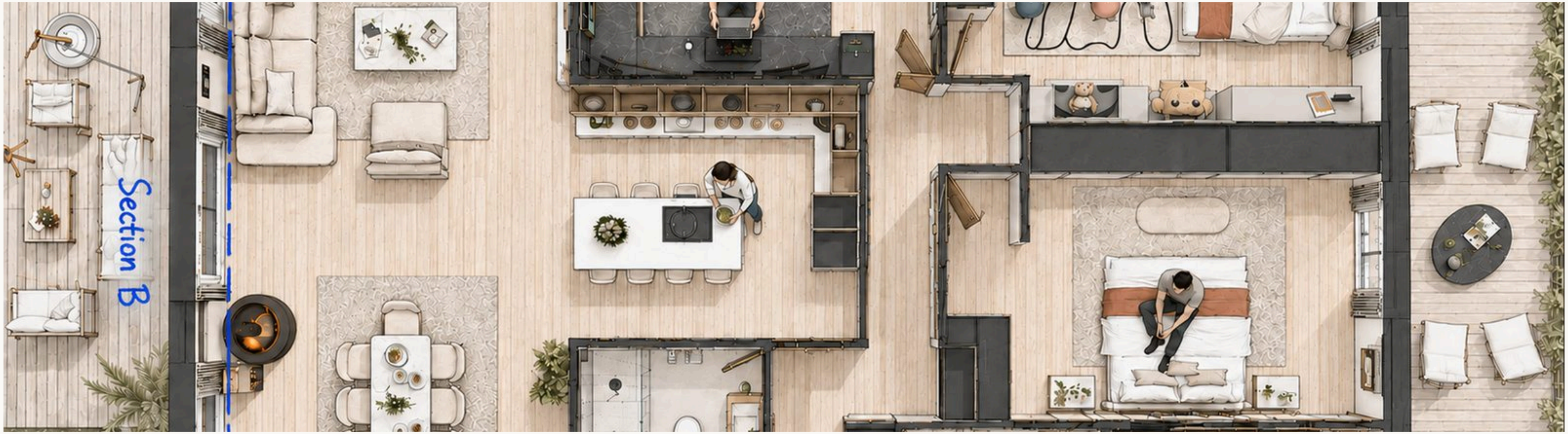
Floor Plan

## D — Selected

Highest quality, highest cost — complete architectural revision

Master Plan

Floor Plan



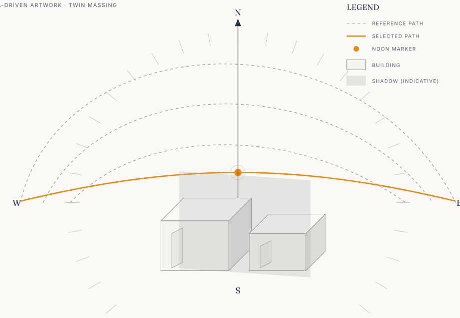

Illustrated floor plan — final proposal.

Section A-A — final proposal, illustrated.

Section B-B — final proposal, illustrated.

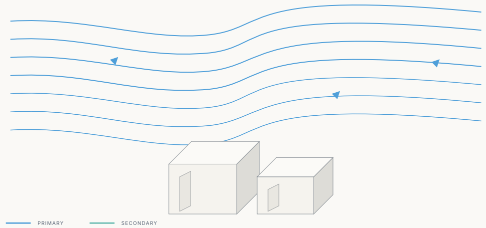
### Solar Path Diagram

DATA-DRIVEN ARTWORK · TWIN MASSING




### Architectural Wind Flow

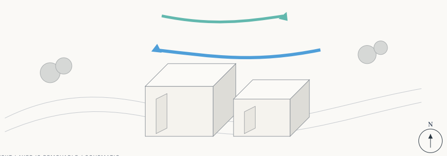
QUALITATIVE STREAMLINE SYSTEM · TWIN MASSING




| DAYLIGHT | NOON ALTITUDE | SUNRISE | SUNSET |
|---|---|---|---|
| 8.0 h | 14.4° | 08:23 | 16:20 |

### Wind Exposure

DIRECTIONAL EXPOSURE · TWIN MASSING

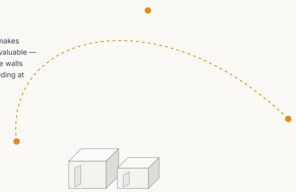



CONTEXT LAYER IS REMOVABLE / SCHEMATIC

### Solar Architectural Reading

INTERPRETIVE READING

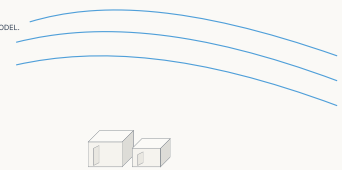
Very low midday solar altitude makes unobstructed southern access valuable — nearby massing, planting, or site walls can meaningfully shade the building at this time of year.



### Wind Envelope Reading

INTERPRETIVE READING

Reported speed: 6.50 M/S · MODEL.



### PREVAILING SECTOR

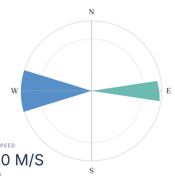
PRIMARY → SECONDARY

PRIMARY DIRECTION
W / SW

SECONDARY DIRECTION
E / NE

SEASON / STATE
WINTER

FREQUENCY STATUS
NOT ESTABLISHED




WIND SPEED
6.50 M/S
· MODEL

### ENVIRONMENTAL DATA DISCLOSURE

Important information about this data

FREQUENCY
frequency not established

WIND SPEED
Modeled wind speed

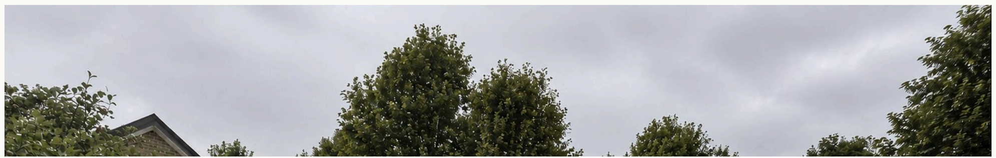
DIRECTION
Qualitative directional interpretation

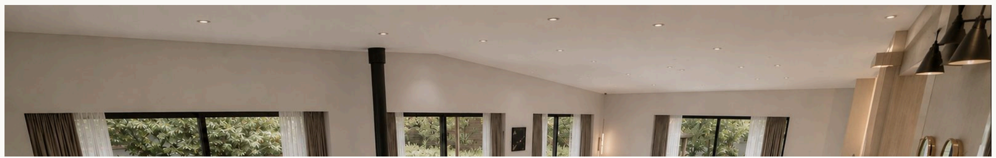
SOURCE DATA
Seasonal, not monthly

The supplied report does not establish verified monthly directional frequency for this site. This diagram visualizes the approved seasonal exposure only — it does not imply precise measurement.

VISUALIZATION

Exterior — View 03





RESIDENTIAL — LUXURY COASTAL VILLA, NEW CONSTRUCTION

# Villa Efe

Villa Efe is a ground-up luxury villa on a steep Mediterranean waterfront site in Northern Cyprus, organizing four privacy-graded levels — from a below-grade service basement to a rooftop leisure deck — around uninterrupted sea views, engineered coastal excavation, and a continuous indoor-outdoor living sequence.

Architectural Design, Visualization, Interior & Landscape Coordination
3ds Max / V-Ray / Corona Renderer / AutoCAD / Revit

PROCESS

## 01 — CLIENT CHALLENGE

*What did the client need, and why did it require starting from scratch?*

An outdated existing residence occupied the site, but the client's spatial requirements exceeded what that building could accommodate, making renovation impractical. The brief called for a contemporary luxury villa maximizing the site's sea views while providing complete privacy, generous family spaces, premium leisure facilities, and long-term functionality — with a clear split between public and private areas, independent staff accommodation, large parking capacity, and an indoor-outdoor living experience throughout.

## 02 — SITE

*What was the physical starting condition?*

The site is a rare waterfront plot directly at sea level with uninterrupted panoramic Mediterranean views, on land with a very steep natural slope that creates multiple access levels. Standing inside the villa facing the sea means facing true north; the opposite side of the site opens onto mountain and forest views. The same site also brought real constraints: extreme proximity to the sea, a high groundwater level, and significant level differences across the plot.

## 03 — CONSTRAINTS

*What limits shaped every decision from here on?*

Basement excavation this close to the sea required specialized engineering — soil retention systems and structural concrete techniques suited to coastal construction were built into the design from the earliest stage. Beyond the engineering, every level had to maximize sea views from its major living spaces while still preserving privacy, and the steep, groundwater-affected site had to absorb a full below-grade basement without compromising spatial clarity above it.

## 04 — DESIGN THINKING

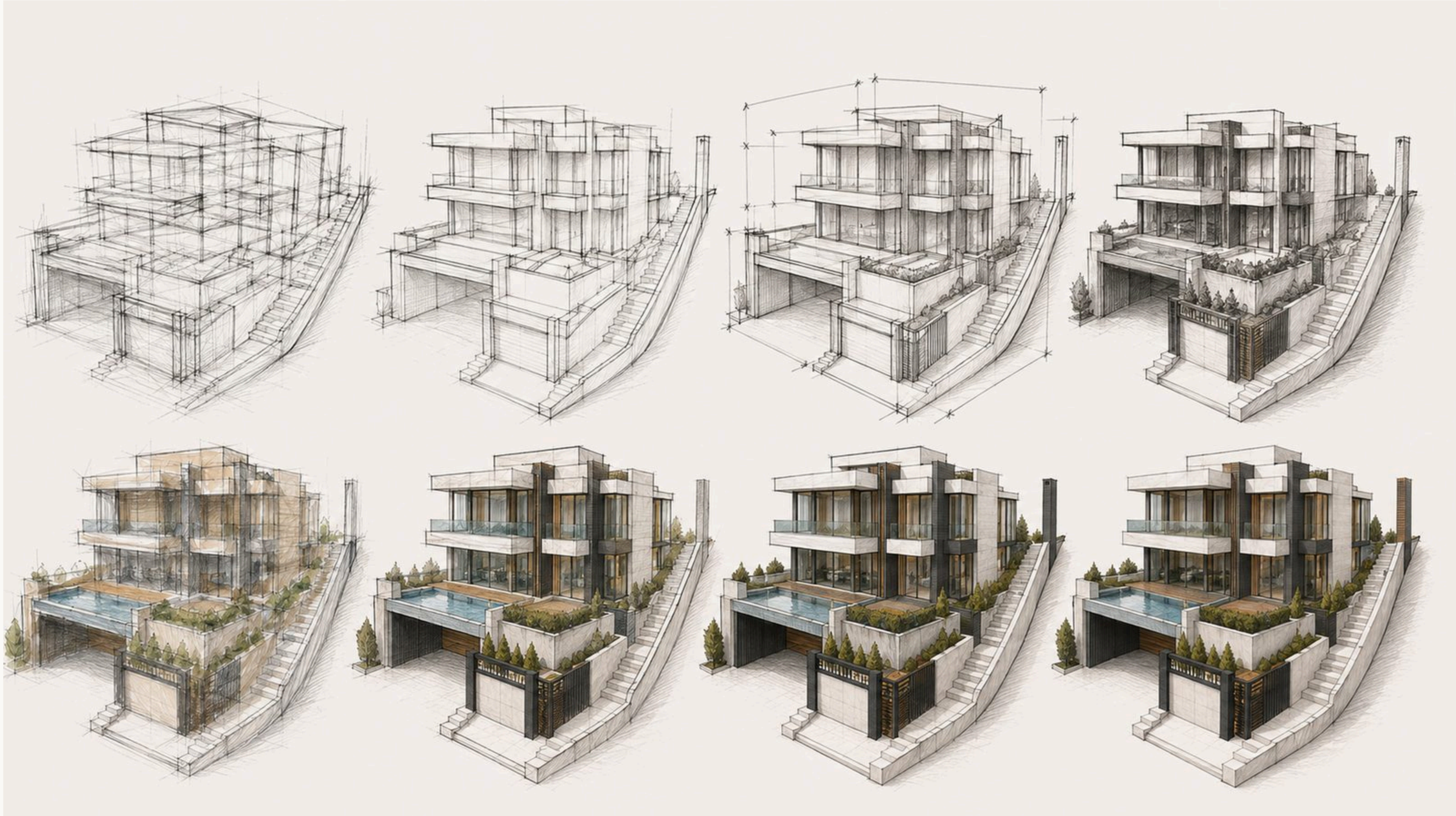
*How was the problem approached?*

Rather than a collection of isolated rooms, the villa was conceived as a sequence of interconnected living environments, with interior and exterior spaces continuously interacting through large openings, terraces, water features, and framed views. The building is organized vertically by privacy level — service and staff space in the basement, public family life on the ground floor, private bedrooms on the first floor, and leisure space on the roof — connected by a central stair and elevator, and lit throughout by a multi-story interior terrarium that carries daylight from the ground floor to the roof.

## 05 — FINAL DECISION

*How did the final design come together?*

The design evolved over roughly eight months through continuous collaboration between the design team and the homeowners, with regular meetings to evaluate every important decision before approval. The owners actively participated in space planning, interior layouts, room relationships, materials, color palettes, furniture concepts, and overall atmosphere, with multiple alternatives presented throughout the



Design development — from early massing studies to final detailing.

### Basement

Service, entertainment & staff accommodation

### Ground Floor

Public family & social spaces

### First Floor

Private bedrooms

### Roof Terrace

Leisure & panoramic outdoor living

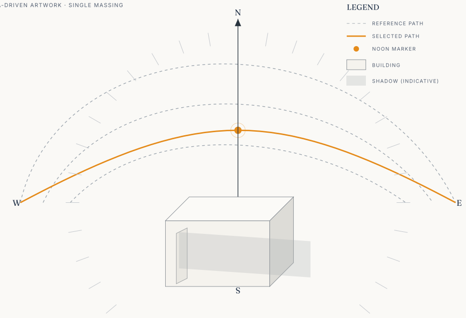

Site plan with section cut-lines.

Section A-A — illustrated.

Section B-B — illustrated.

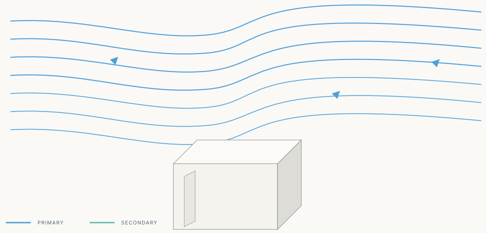
## Solar Path Diagram

DATA-DRIVEN ARTWORK · SINGLE MASSING




## Architectural Wind Flow

QUALITATIVE STREAMLINE SYSTEM · SINGLE MASSING




DAYLIGHT
10.1 h

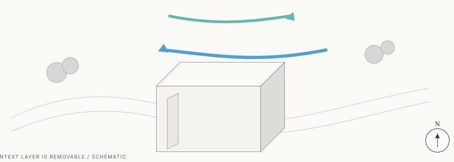
NOON ALTITUDE
34.6°

SUNRISE
06:53

SUNSET
17:02

## Wind Exposure

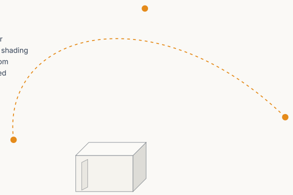
DIRECTIONAL EXPOSURE · SINGLE MASSING




## Solar Architectural Reading

INTERPRETIVE READING

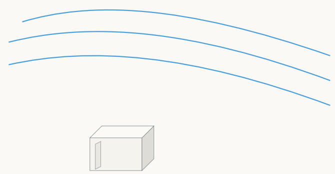
The winter transition moves solar altitude quickly month to month; shading and glazing strategies benefit from flexibility rather than a single fixed assumption.



## Wind Envelope Reading

INTERPRETIVE READING

Reported speed: 3.0 M/S.



### PREVAILING SECTOR

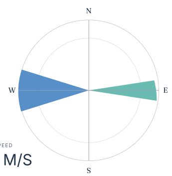
PRIMARY → SECONDARY

PRIMARY DIRECTION
W / NW

SECONDARY DIRECTION
E / SE

SEASON / STATE
WINTER

FREQUENCY STATUS
NOT ESTABLISHED




WIND SPEED
3.0 M/S

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3.0 M/S

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VISUALIZATION

Exterior — View 03